Payroll Cost Optimization & Salary Fairness Analyzer

Converted from the uploaded Jupyter Notebook (.ipynb), including saved chart/image outputs where available.

Code Cell 1

```
#1.Import Required Libraries
import pandas as pd
import numpy as np
print("Payroll Project Started")
print("Pandas version:",
     pd.__version__)
print("Numpy version:",
     np.__version__)
```

Output

```
Payroll Project Started
Pandas version: 2.3.3
Numpy version: 2.1.3
```

Code Cell 2

```
#2.Create employeee Payroll Dataset
import pandas as pd
import numpy as np
np.random.seed(42)
n=100000
data={
    "Employee_ID": range(1,n+1),
    "Department": np.random.choice(
        ["HR","Finance","IT","Sales","Marketing"],
        n
    ),
    "Gender": np.random.choice(
        ["Male","Female"],
        n
    ),
    "Age": np.random.randint(22,61,n),
    "Experience_Yers":
    np.random.randint(0,36,n),
    "job_Level": np.random.choice(
        ["Junior","Mid","Senior","Manager"],
        n
    ),
    "Basic_Salary":
    np.random.randint(18000,100001,n),
    "Allowance":
    np.random.randint(1000,10001,n),
    "Bonus": np.random.randint(0,20001,n),
    "Overtime_Hours":
    np.random.randint(0,41,n),
    "Performance_Score":
    np.random.randint(1,6,n),
    "Education": np.random.choice(
        ["Bachelor","Master","PhD"],
        n
    ),
    "Location":np.random.choice(
        ["Chennai","Bangalore","Coimbatore","Hyderabad"],
        n
    )
}
df=pd.DataFrame(data)
df["Total_Salary"]=(
    df["Basic_Salary"]+
    df["Allowance"] + df["Bonus"]
)
print("Dataset created successfully!")
print("Rows:",df.shape[0])
print("Columns:",df.shape[1])
df.head()
```

Output

```
Dataset created successfully!
Rows: 100000
Columns: 14
```

Output

```
   Employee_ID Department  Gender  Age  Experience_Yers job_Level  \
0            1      Sales  Female   58                8       Mid
1            2  Marketing  Female   31               32   Manager
2            3         IT  Female   42               18    Senior
3            4  Marketing    Male   28               15   Manager
4            5  Marketing    Male   50                7       Mid

   Basic_Salary  Allowance  Bonus  Overtime_Hours  Performance_Score  \
0         42924       7886   4430              14                  5
1         27995       5673  15074              20                  3
2         22902       7749   8835              25                  2
3         54682       1452   1160              38                  3
4         37899       8996  19619              11                  2

  Education   Location  Total_Salary
0       PhD  Hyderabad         55240
1  Bachelor  Hyderabad         48742
2       PhD  Hyderabad         39486
3    Master  Hyderabad         57294
4  Bachelor  Bangalore         66514
```

Code Cell 3

```
#3.Save Dataset as CSV File
df.to_csv("payroll_dataset.csv",index=False)
print("Dataset saved successfully!")
print("File Name: payroll_dataset.csv")
print("Dataset Shape:",df.shape)
```

Output

```
Dataset saved successfully!
File Name: payroll_dataset.csv
Dataset Shape: (100000, 14)
```

Code Cell 4

```
#4.Data Inspection and Quality Check
print("First 5 Employee Records:")
display(df.head())
print("\nDataset Shape:")
print(df.shape)
print("\nColumn Names:")
print(df.columns.tolist())
print("\nData Types:")
print(df.dtypes)
print("\nMissing Values:")
print(df.isnull().sum())
print("\nDuplicate Rows:")
print(df.duplicated().sum())
```

Output

```
First 5 Employee Records:
```

Output

```
   Employee_ID Department  Gender  Age  Experience_Yers job_Level  \
0            1      Sales  Female   58                8       Mid
1            2  Marketing  Female   31               32   Manager
2            3         IT  Female   42               18    Senior
3            4  Marketing    Male   28               15   Manager
4            5  Marketing    Male   50                7       Mid

   Basic_Salary  Allowance  Bonus  Overtime_Hours  Performance_Score  \
0         42924       7886   4430              14                  5
1         27995       5673  15074              20                  3
2         22902       7749   8835              25                  2
3         54682       1452   1160              38                  3
4         37899       8996  19619              11                  2

  Education   Location  Total_Salary
0       PhD  Hyderabad         55240
1  Bachelor  Hyderabad         48742
2       PhD  Hyderabad         39486
3    Master  Hyderabad         57294
4  Bachelor  Bangalore         66514
```

Output

```
Dataset Shape:
(100000, 14)
```

```
Column Names:
['Employee_ID', 'Department', 'Gender', 'Age', 'Experience_Yers', 'job_Level', 'Basic_Salary', 'Allowance',
'Bonus', 'Overtime_Hours', 'Performance_Score', 'Education', 'Location', 'Total_Salary']

Data Types:
Employee_ID           int64
Department           object
Gender               object
Age                   int64
Experience_Yers       int64
job_Level            object
Basic_Salary          int64
Allowance             int64
Bonus                 int64
Overtime_Hours        int64
Performance_Score     int64
Education            object
Location             object
Total_Salary          int64
dtype: object

Missing Values:
Employee_ID          0
Department           0
Gender               0
Age                  0
Experience_Yers      0
job_Level            0
Basic_Salary         0
Allowance            0
Bonus                0
Overtime_Hours       0
Performance_Score    0
Education            0
Location             0
Total_Salary         0
dtype: int64

Duplicate Rows:
0
```

Code Cell 5

```
#5.Department-wise Payroll cost Analysis
department_analysis = df.groupby("Department").agg(
    Employee_Count=("Employee_ID", "count"),
    Total_Payroll_Cost=("Total_Salary", "sum"),
    Average_Salary=("Total_Salary", "mean")
).reset_index()
department_analysis = department_analysis.sort_values(
    by="Total_Payroll_Cost",
    ascending=False
)
print("Department-wise Payroll Analysis")
display(department_analysis.round(2))
```

Output

```
Department-wise Payroll Analysis
```

Output

	Department	Employee_Count	Total_Payroll_Cost	Average_Salary
3	Marketing	20187	1506633126	74633.83
0	Finance	20082	1495331636	74461.29
1	HR	20018	1487263641	74296.32
4	Sales	19981	1484211166	74281.13
2	IT	19732	1476075672	74806.19

Code Cell 6

```
#6.Department-wisw Payroll cost Visualization
import matplotlib.pyplot as plt

# Department-wise Payroll Cost Chart

plt.figure(figsize=(10, 6))

plt.bar(
    department_analysis["Department"],
    department_analysis["Total_Payroll_Cost"]
)

plt.title("Department-wise Total Payroll Cost")
```

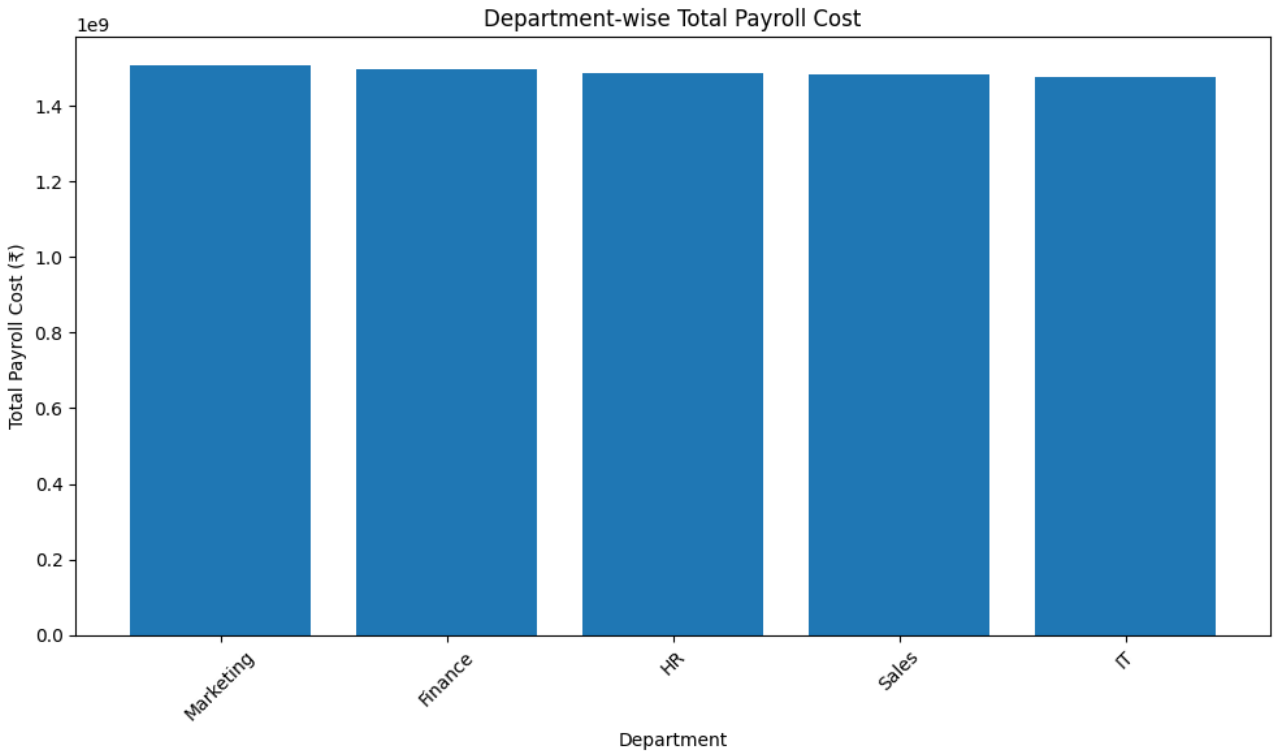
```
plt.xlabel("Department")
plt.ylabel("Total Payroll Cost (■)")
plt.xticks(rotation=45)

plt.tight_layout()
plt.show()
```

Output

```
<Figure size 1000x600 with 1 Axes>
```

Chart / Image Output



Code Cell 7

```
#7.Gender-wise Salary Analysis

gender_analysis = df.groupby("Gender").agg(
    Employee_Count=("Employee_ID", "count"),
    Average_Basic_Salary=("Basic_Salary", "mean"),
    Average_Total_Salary=("Total_Salary", "mean")
).reset_index()

print("Gender-wise Salary Comparison")
display(gender_analysis.round(2))
```

Output

```
Gender-wise Salary Comparison
```

Output

	Gender	Employee_Count	Average_Basic_Salary	Average_Total_Salary
0	Female	50099	59042.16	74559.73
1	Male	49901	58933.58	74430.32

Code Cell 8

```
#7.1 Calculate average total salary difference

gender_salary = df.groupby("Gender")["Total_Salary"].mean()

male_avg = gender_salary.get("Male", 0)
female_avg = gender_salary.get("Female", 0)

print("Male Average Salary:", round(male_avg, 2))
print("Female Average Salary:", round(female_avg, 2))
print("Salary Difference:", round(abs(male_avg - female_avg), 2))
```

Output

```
Male Average Salary: 74430.32
Female Average Salary: 74559.73
Salary Difference: 129.41
```

Code Cell 9

```
# 8. Job-Level-wise Salary Analysis
job_analysis = df.groupby("job_Level").agg(
    Employee_Count=("Employee_ID", "count"),
    Average_Basic_Salary=("Basic_Salary", "mean"),
    Average_Total_Salary=("Total_Salary", "mean"),
    Total_Payroll_Cost=("Total_Salary", "sum")
).reset_index()

job_analysis = job_analysis.sort_values(
    by="Average_Total_Salary",
    ascending=False
)
print("Job-Level-wise Salary Analysis")
display(job_analysis.round(2))
```

Output

```
Job-Level-wise Salary Analysis
```

Output

```
  job_Level  Employee_Count  Average_Basic_Salary  Average_Total_Salary  \
0    Junior           24818              59240.93              74809.95
2       Mid           24950              59014.07              74490.05
1   Manager           25002              58986.95              74476.59
3    Senior           25230              58714.35              74208.94

   Total_Payroll_Cost
0          1856633240
2          1858526769
1          1862063679
3          1872291553
```

Code Cell 10

```
# 9. Experience-wise Salary Analysis

# Check and standardize column names
df.columns = df.columns.str.strip()

# Find the actual experience column name
experience_col = [
    col for col in df.columns
    if "experience" in col.lower()
][0]

print("Experience column found:", experience_col)

# Perform salary analysis
experience_analysis = df.groupby(experience_col).agg(
    Employee_Count=("Employee_ID", "count"),
    Average_Basic_Salary=("Basic_Salary", "mean"),
    Average_Total_Salary=("Total_Salary", "mean")
).reset_index()

experience_analysis = experience_analysis.sort_values(
    by=experience_col
)

print("Experience-wise Salary Analysis")
display(experience_analysis.round(2))
```

Output

```
Experience column found: Experience_Yers
Experience-wise Salary Analysis
```

Output

```
    Experience_Yers  Employee_Count  Average_Basic_Salary  \
0                 0            2800              59463.93
1                 1            2792              57931.46
2                 2            2789              58125.87
3                 3            2760              59503.67
4                 4            2862              59471.35
```

```
5                5           2807             58588.09
6                6           2714             58425.44
7                7           2827             59112.72
8                8           2731             58999.61
9                9           2768             58974.88
10              10           2836             59355.46
11              11           2715             58944.83
12              12           2754             59488.31
13              13           2714             59115.63
14              14           2762             58921.19
15              15           2804             59000.88
16              16           2835             59160.11
17              17           2690             58598.74
18              18           2800             59235.54
19              19           2830             58400.52
20              20           2755             59022.23
21              21           2775             58890.62
22              22           2863             59311.68
23              23           2775             58928.80
24              24           2754             59323.19
25              25           2727             58359.88
26              26           2846             59293.65
27              27           2741             59380.92
28              28           2788             58932.49
29              29           2767             59052.14
30              30           2728             58996.34
31              31           2813             59354.57
32              32           2745             59040.63
33              33           2784             59343.39
34              34           2774             58595.87
35              35           2775             58871.10

    Average_Total_Salary
0               74962.00
1               73418.79
2               73614.10
3               75196.51
4               75024.03
5               74092.19
6               73962.92
7               74729.59
8               74442.90
9               74545.47
10              74794.96
11              74636.53
12              74908.65
13              74409.17
14              74435.45
15              74381.73
16              74663.57
17              74086.83
18              74758.80
19              74157.72
20              74545.35
21              74394.12
22              74679.12
23              74621.85
24              74770.76
25              73923.00
26              75010.13
27              74658.35
28              74391.55
29              74373.09
30              74633.58
31              74800.43
32              74449.63
33              74875.73
34              74034.47
35              74383.05
```

Code Cell 11

```
# 10. Experience vs Average Salary Chart

import matplotlib.pyplot as plt

# Find the correct experience column
experience_col = [
    col for col in df.columns
    if "experience" in col.lower()
][0]

# Calculate average salary by experience
experience_analysis = df.groupby(experience_col).agg(
    Average_Total_Salary=("Total_Salary", "mean")
```

```
).reset_index()

experience_analysis = experience_analysis.sort_values(
    by=experience_col
)

# Create the chart
plt.figure(figsize=(10, 5))

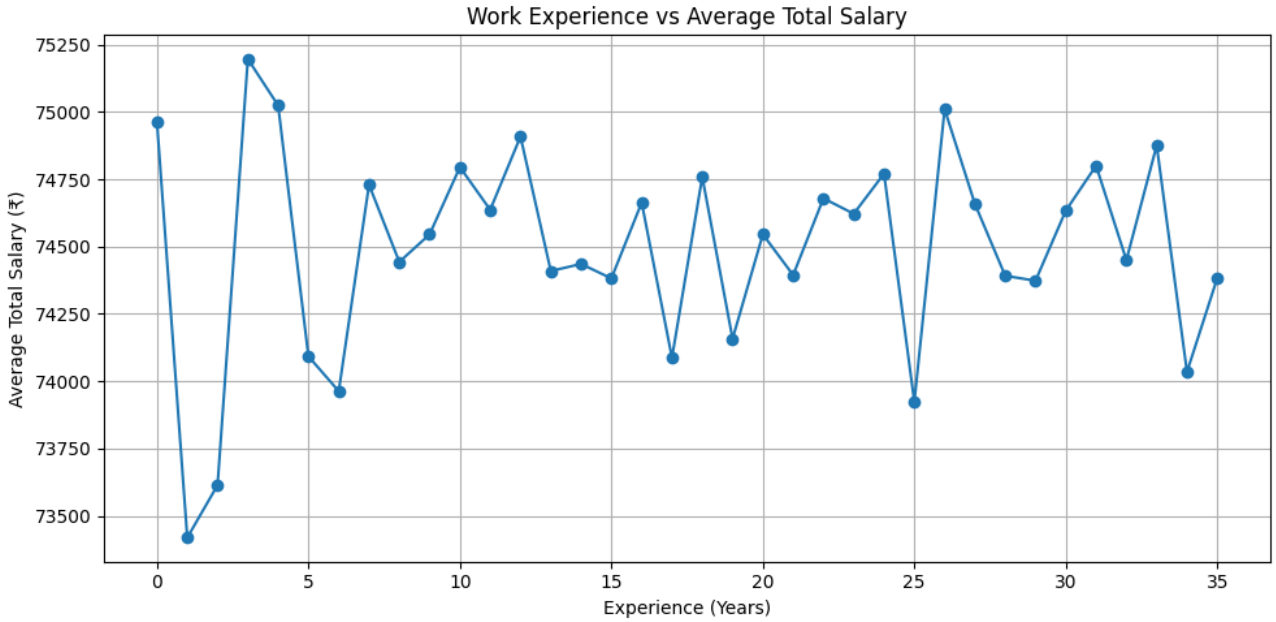
plt.plot(
    experience_analysis[experience_col],
    experience_analysis["Average_Total_Salary"],
    marker="o"
)

plt.title("Work Experience vs Average Total Salary")
plt.xlabel("Experience (Years)")
plt.ylabel("Average Total Salary (■)")
plt.grid(True)
plt.tight_layout()
plt.show()
```

Output

```
<Figure size 1000x500 with 1 Axes>
```

Chart / Image Output



Code Cell 12

```
# 11. Department-wise Payroll Budget Analysis

department_budget = df.groupby("Department").agg(
    Employee_Count=("Employee_ID", "count"),
    Total_Payroll_Cost=("Total_Salary", "sum"),
    Average_Salary=("Total_Salary", "mean")
).reset_index()

# Calculate percentage of total payroll
total_payroll = department_budget["Total_Payroll_Cost"].sum()

department_budget["Budget_Percentage"] = (
    department_budget["Total_Payroll_Cost"]
    / total_payroll * 100
)

# Sort departments by payroll cost
department_budget = department_budget.sort_values(
    by="Total_Payroll_Cost",
    ascending=False
)

print("Department-wise Payroll Budget Analysis")
display(department_budget.round(2))
```

```
print("\nTotal Company Payroll Cost: ■", round(total_payroll, 2))
```

Output

```
Department-wise Payroll Budget Analysis
```

Output

```
  Department  Employee_Count  Total_Payroll_Cost  Average_Salary  \
3  Marketing           20187          1506633126        74633.83
0    Finance           20082          1495331636        74461.29
1         HR           20018          1487263641        74296.32
4      Sales           19981          1484211166        74281.13
2         IT           19732          1476075672        74806.19

   Budget_Percentage
3              20.22
0              20.07
1              19.96
4              19.92
2              19.81
```

Output

```
Total Company Payroll Cost: ■ 7449515241
```

Code Cell 13

```
#12. Department-wise Payroll Budget Chart
import matplotlib.pyplot as plt

plt.figure(figsize=(10, 6))

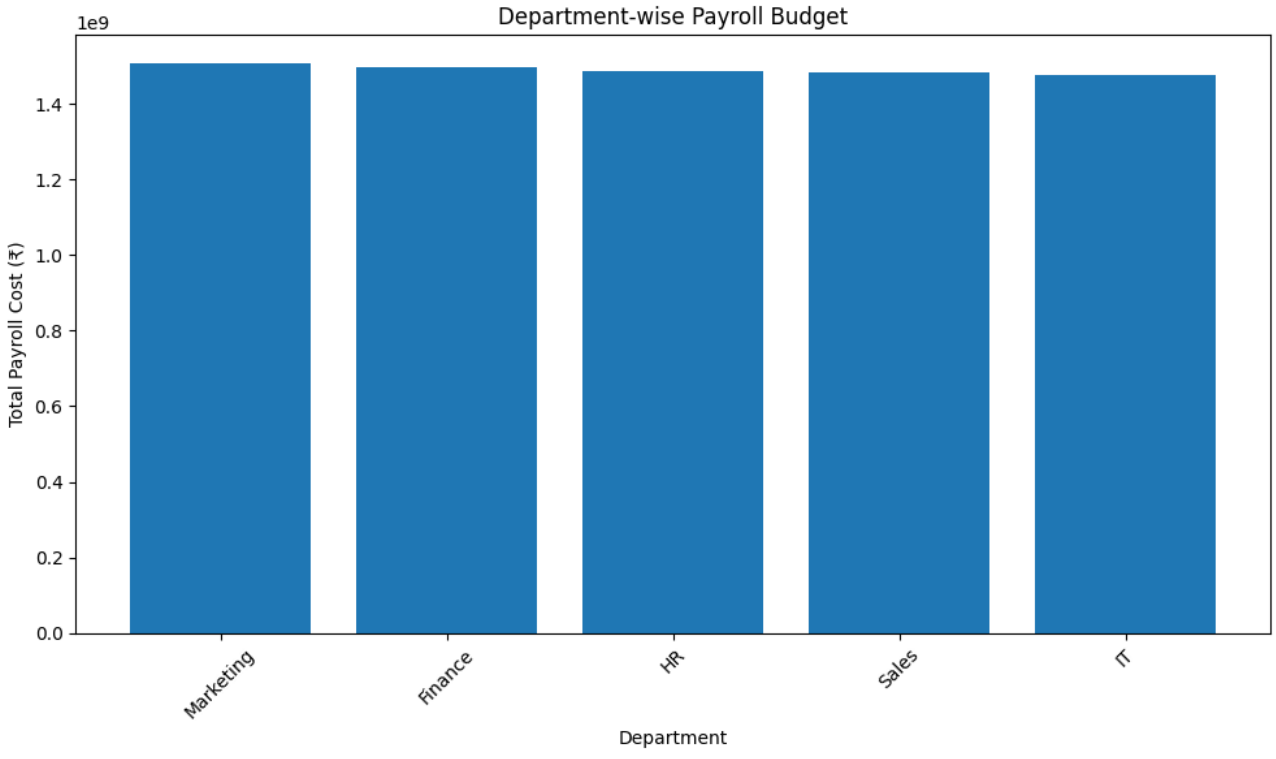
plt.bar(
    department_budget["Department"],
    department_budget["Total_Payroll_Cost"]
)

plt.title("Department-wise Payroll Budget")
plt.xlabel("Department")
plt.ylabel("Total Payroll Cost (■)")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

Output

```
<Figure size 1000x600 with 1 Axes>
```

Chart / Image Output



Code Cell 14

```
#13 Salary Fairness Analysis by Department and Gender
# Department and Gender Salary Comparison

fairness_analysis = df.groupby(
    ["Department", "Gender"]
).agg(
    Employee_Count=("Employee_ID", "count"),
    Average_Salary=("Total_Salary", "mean")
).reset_index()

print("Salary Fairness Analysis by Department and Gender")
display(fairness_analysis.round(2))
```

Output

```
Salary Fairness Analysis by Department and Gender
```

Output

```
  Department  Gender  Employee_Count  Average_Salary
0    Finance  Female           10186        74604.64
1    Finance    Male            9896        74313.75
2         HR  Female           10085        74495.03
3         HR    Male            9933        74094.56
4         IT  Female            9916        74665.48
5         IT    Male            9816        74948.33
6  Marketing  Female            9958        74523.97
7  Marketing    Male           10229        74740.78
8      Sales  Female            9954        74509.77
9      Sales    Male           10027        74054.15
```

Code Cell 15

```
#13.1# Compare average salaries by gender within each department

fairness_pivot = fairness_analysis.pivot(
    index="Department",
    columns="Gender",
    values="Average_Salary"
)

fairness_pivot["Salary_Difference"] = (
    fairness_pivot["Male"] - fairness_pivot["Female"]
).abs()

print("Department-wise Gender Salary Difference")
display(fairness_pivot.round(2))
```

Output

```
Department-wise Gender Salary Difference
```

Output

```
Gender        Female      Male  Salary_Difference
Department
Finance     74604.64  74313.75             290.89
HR          74495.03  74094.56             400.46
IT          74665.48  74948.33             282.84
Marketing   74523.97  74740.78             216.82
Sales       74509.77  74054.15             455.62
```

Code Cell 16

```
# 14. Key Business Insights and Recommendations

print("=" * 55)
print("PAYROLL BUSINESS INSIGHTS")
print("=" * 55)

# 1. Department with highest payroll cost
highest_cost = department_budget.iloc[0]

print("\n1. Highest Payroll Cost Department")
print("Department:", highest_cost["Department"])
print("Total Payroll Cost: ■",
      round(highest_cost["Total_Payroll_Cost"], 2))

# 2. Department with the most employees
most_employees = department_budget.loc[
    department_budget["Employee_Count"].idxmax()
```

```
]

print("\n2. Department with Most Employees")
print("Department:", most_employees["Department"])
print("Employee Count:", most_employees["Employee_Count"])

# 3. Department with highest average salary
highest_average = department_budget.loc[
    department_budget["Average_Salary"].idxmax()
]

print("\n3. Highest Average Salary Department")
print("Department:", highest_average["Department"])
print("Average Salary: ■",
      round(highest_average["Average_Salary"], 2))

# 4. Budget share of the highest-cost department
print("\n4. Payroll Budget Share")
print("Budget Percentage:",
      round(highest_cost["Budget_Percentage"], 2), "%")

print("\n5. Recommendations")
print("- Review overtime and bonus expenses.")
print("- Compare salary differences by job level and experience.")
print("- Investigate unusual payroll costs before making changes.")
print("- Do not reduce salaries based on department cost alone.")
```

Output

```
======================================================
PAYROLL BUSINESS INSIGHTS
======================================================

1. Highest Payroll Cost Department
Department: Marketing
Total Payroll Cost: ■ 1506633126

2. Department with Most Employees
Department: Marketing
Employee Count: 20187

3. Highest Average Salary Department
Department: IT
Average Salary: ■ 74806.19

4. Payroll Budget Share
Budget Percentage: 20.22 %

5. Recommendations
- Review overtime and bonus expenses.
- Compare salary differences by job level and experience.
- Investigate unusual payroll costs before making changes.
- Do not reduce salaries based on department cost alone.
```

Code Cell 17

```
# Step 15. Experience-wise Salary Analysis - Error Fix

import matplotlib.pyplot as plt

# Check actual column names
print("Experience Analysis Columns:")
print(experience_analysis.columns.tolist())

# Automatically find the average salary column
salary_col = [
    col for col in experience_analysis.columns
    if "average" in col.lower() and "salary" in col.lower()
][0]

plt.figure(figsize=(10, 5))

plt.plot(
    experience_analysis[experience_col],
    experience_analysis[salary_col],
    marker="o"
)

plt.title("Experience vs Average Total Salary")
plt.xlabel("Experience (Years)")
plt.ylabel("Average Salary (■)")
plt.grid(True)
plt.tight_layout()
```

```
plt.show()
```

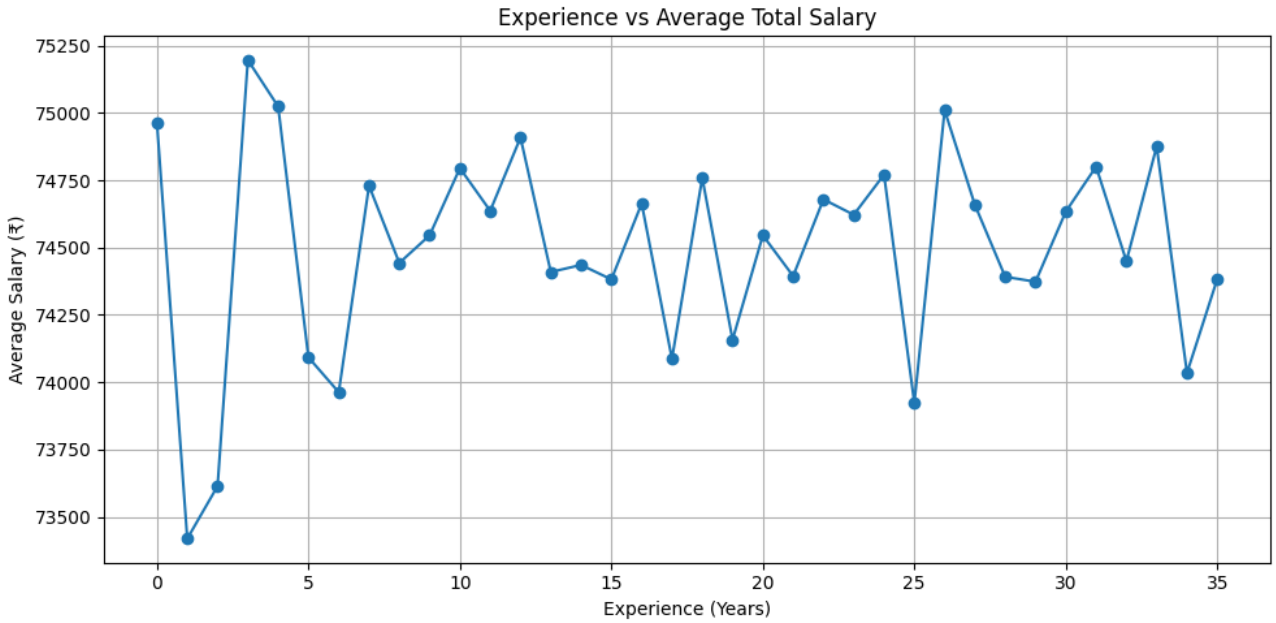
Output

```
Experience Analysis Columns:
['Experience_Yers', 'Average_Total_Salary']
```

Output

```
<Figure size 1000x500 with 1 Axes>
```

Chart / Image Output



Code Cell 18

```
# Step 16. Payroll Cost Optimization Analysis

import pandas as pd

# Department-wise payroll cost analysis
cost_analysis = df.groupby("Department").agg(
    Employee_Count=("Employee_ID", "count"),
    Total_Payroll=("Total_Salary", "sum"),
    Average_Salary=("Total_Salary", "mean"),
    Total_Bonus=("Bonus", "sum"),
    Total_Allowance=("Allowance", "sum"),
    Total_Overtime_Hours=("Overtime_Hours", "sum")
).reset_index()

# Calculate percentage of total payroll
total_payroll = cost_analysis["Total_Payroll"].sum()

cost_analysis["Payroll_Percentage"] = (
    cost_analysis["Total_Payroll"] / total_payroll * 100
)

# Display highest-cost departments first
cost_analysis = cost_analysis.sort_values(
    by="Total_Payroll",
    ascending=False
)

print("PAYROLL COST OPTIMIZATION REPORT")
display(cost_analysis.round(2))

print("\nHighest Payroll Cost Department:")
print(cost_analysis.iloc[0]["Department"])

print("\nTotal Company Payroll: ■",
      round(total_payroll, 2))
```

Output

```
PAYROLL COST OPTIMIZATION REPORT
```

Output

```
  Department  Employee_Count  Total_Payroll  Average_Salary  Total_Bonus  \
3  Marketing           20187     1506633126        74633.83    201521848
0    Finance           20082     1495331636        74461.29    200908931
1         HR           20018     1487263641        74296.32    200277783
4      Sales           19981     1484211166        74281.13    199891268
2         IT           19732     1476075672        74806.19    197662548

   Total_Allowance  Total_Overtime_Hours  Payroll_Percentage
3        111418514                402136               20.22
0        110731104                398081               20.07
1        110272509                400306               19.96
4        109654004                397746               19.92
2        108379268                393393               19.81
```

Output

```
Highest Payroll Cost Department:
Marketing

Total Company Payroll: ■ 7449515241
```

Code Cell 19

```
# Step 17. Payroll Visual Dashboard

import matplotlib.pyplot as plt

# Prepare department summary
dashboard_data = df.groupby("Department").agg(
    Total_Payroll=("Total_Salary", "sum"),
    Average_Salary=("Total_Salary", "mean"),
    Employee_Count=("Employee_ID", "count")
).reset_index()

# Chart 1: Department-wise total payroll
plt.figure(figsize=(10, 5))
plt.bar(
    dashboard_data["Department"],
    dashboard_data["Total_Payroll"]
)
plt.title("Department-wise Total Payroll Cost")
plt.xlabel("Department")
plt.ylabel("Total Payroll (■)")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

# Chart 2: Department-wise average salary
plt.figure(figsize=(10, 5))
plt.bar(
    dashboard_data["Department"],
    dashboard_data["Average_Salary"]
)
plt.title("Department-wise Average Salary")
plt.xlabel("Department")
plt.ylabel("Average Salary (■)")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

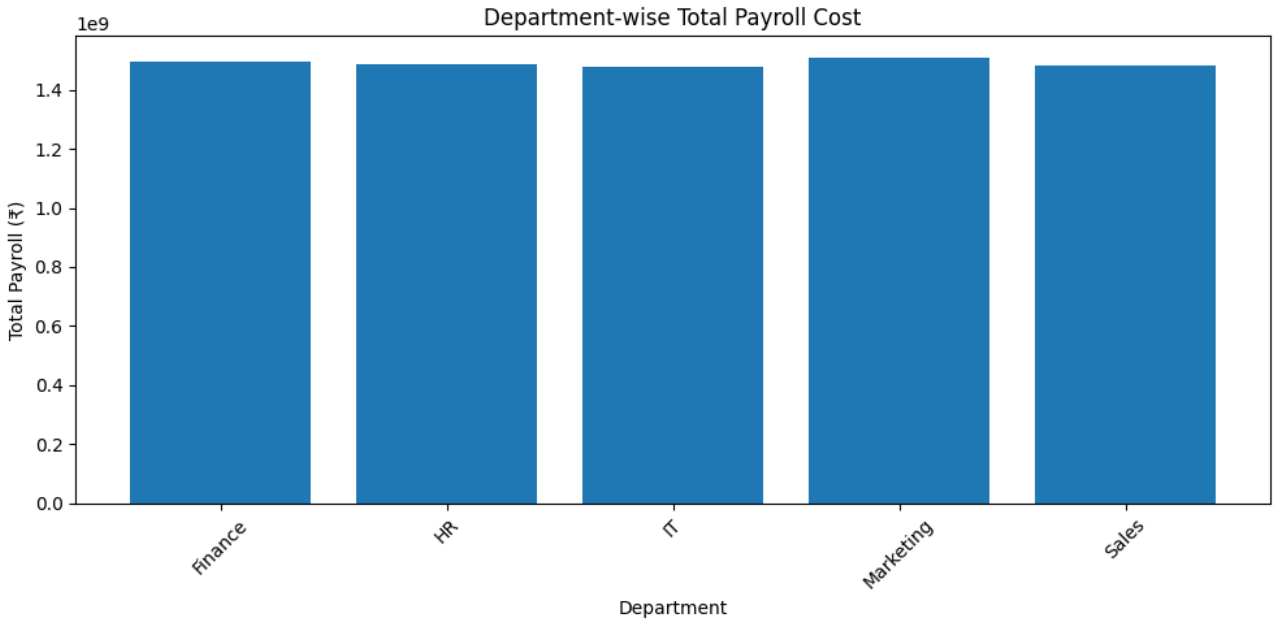
# Chart 3: Employee distribution
plt.figure(figsize=(8, 8))
plt.pie(
    dashboard_data["Employee_Count"],
    labels=dashboard_data["Department"],
    autopct="%1.1f%%"
)
plt.title("Employee Distribution by Department")
plt.show()

print("Dashboard created successfully!")
```

Output

```
<Figure size 1000x500 with 1 Axes>
```

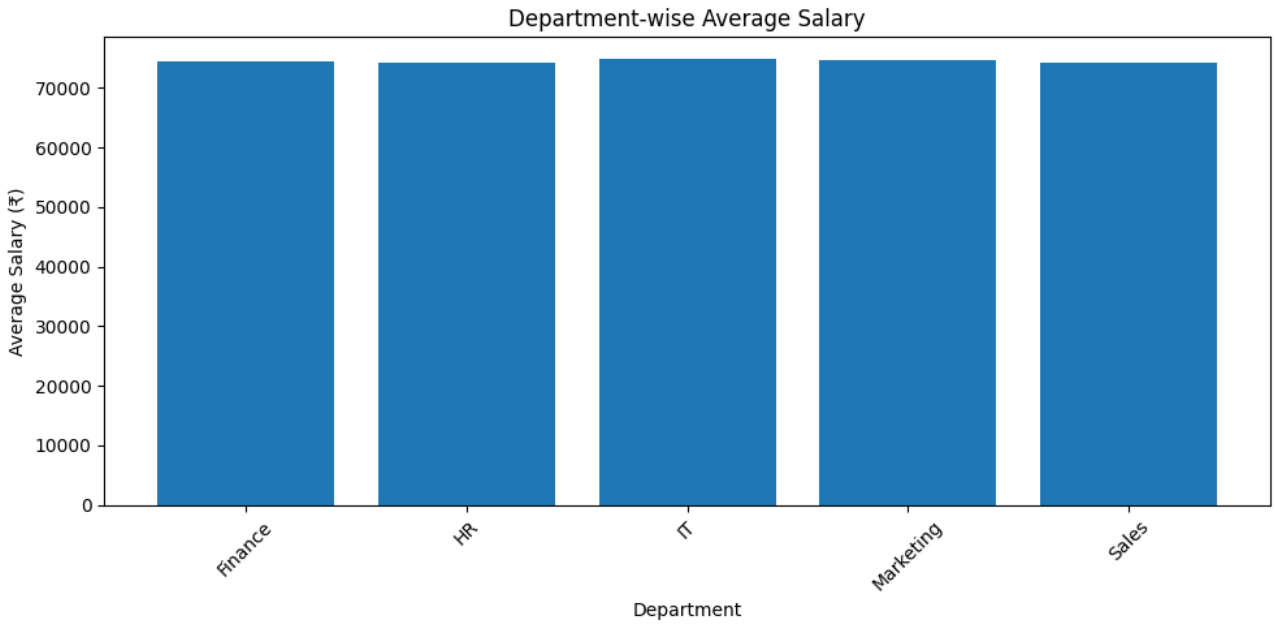
Chart / Image Output



Output

```
<Figure size 1000x500 with 1 Axes>
```

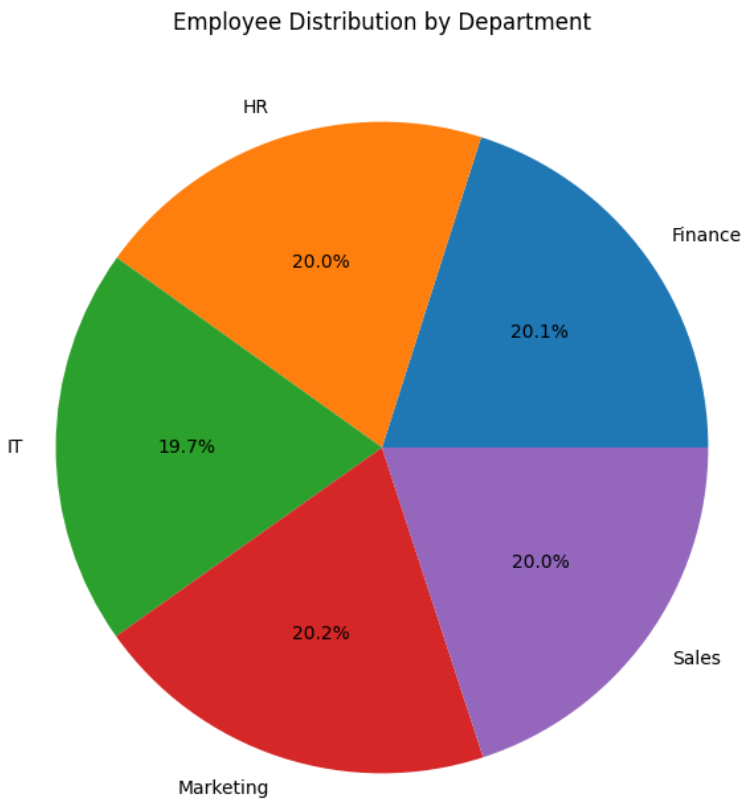
Chart / Image Output



Output

```
<Figure size 800x800 with 1 Axes>
```

Chart / Image Output



Output

```
Dashboard created successfully!
```

Code Cell 20

```
# Step 18. Final Payroll KPI Report

import pandas as pd

print("=" * 50)
print("       FINAL PAYROLL KPI REPORT")
print("=" * 50)

# Overall KPIs
total_employees = df["Employee_ID"].nunique()
total_payroll = df["Total_Salary"].sum()
average_salary = df["Total_Salary"].mean()
median_salary = df["Total_Salary"].median()
total_bonus = df["Bonus"].sum()
total_allowance = df["Allowance"].sum()
total_overtime = df["Overtime_Hours"].sum()
department_count = df["Department"].nunique()

print(f"Total Employees       : {total_employees:,}")
print(f"Number of Departments : {department_count}")
print(f"Total Payroll Cost    : ■{total_payroll:,.2f}")
print(f"Average Salary        : ■{average_salary:,.2f}")
print(f"Median Salary         : ■{median_salary:,.2f}")
print(f"Total Bonus           : ■{total_bonus:,.2f}")
print(f"Total Allowance       : ■{total_allowance:,.2f}")
print(f"Total Overtime Hours  : {total_overtime:,.0f}")

# Department-wise KPI table
department_kpis = df.groupby("Department").agg(
    Employee_Count=("Employee_ID", "nunique"),
    Total_Payroll=("Total_Salary", "sum"),
    Average_Salary=("Total_Salary", "mean")
).reset_index()

department_kpis = department_kpis.sort_values(
    "Total_Payroll", ascending=False
)
```

```
print("\nDEPARTMENT-WISE KPI SUMMARY")
display(department_kpis.round(2))

print("\nKPI report generated successfully!")
```

Output

```
================================================
       FINAL PAYROLL KPI REPORT
================================================
Total Employees       : 100,000
Number of Departments : 5
Total Payroll Cost    : ■7,449,515,241.00
Average Salary        : ■74,495.15
Median Salary         : ■74,575.00
Total Bonus           : ■1,000,262,378.00
Total Allowance       : ■550,455,399.00
Total Overtime Hours  : 1,991,662

DEPARTMENT-WISE KPI SUMMARY
```

Output

	Department	Employee_Count	Total_Payroll	Average_Salary
3	Marketing	20187	1506633126	74633.83
0	Finance	20082	1495331636	74461.29
1	HR	20018	1487263641	74296.32
4	Sales	19981	1484211166	74281.13
2	IT	19732	1476075672	74806.19

Output

```
KPI report generated successfully!
```

Code Cell 21

```
# Step 19. Final Business Recommendations

print("=" * 60)
print("FINAL BUSINESS RECOMMENDATIONS")
print("=" * 60)

# 1. Highest payroll cost department
dept_cost = df.groupby("Department")["Total_Salary"].sum()
highest_cost_dept = dept_cost.idxmax()

print("\n1. Payroll Cost Review")
print(f"Highest payroll cost department: {highest_cost_dept}")
print("Recommendation: Review staffing, workload and payroll expenses.")

# 2. Overtime analysis
overtime_dept = df.groupby("Department")["Overtime_Hours"].mean()
highest_overtime_dept = overtime_dept.idxmax()

print("\n2. Overtime Management")
print(f"Highest average overtime department: {highest_overtime_dept}")
print("Recommendation: Check workload distribution and overtime patterns.")

# 3. Bonus analysis
bonus_dept = df.groupby("Department")["Bonus"].mean()
highest_bonus_dept = bonus_dept.idxmax()

print("\n3. Bonus Analysis")
print(f"Highest average bonus department: {highest_bonus_dept}")
print("Recommendation: Review bonus criteria and performance policies.")

# 4. Salary fairness
gender_salary = df.groupby("Gender")["Total_Salary"].mean()

print("\n4. Salary Fairness")
display(gender_salary.round(2))
print("Recommendation: Compare salaries by job level, experience and performance.")

# 5. Final suggestions
print("\n5. Overall Recommendations")
print("- Monitor department-wise payroll costs regularly.")
print("- Investigate unusual salary and overtime patterns.")
print("- Use transparent salary and bonus policies.")
print("- Make cost decisions without compromising employee fairness.")

print("\nBusiness recommendations generated successfully!")
```

Output

```
============================================================
FINAL BUSINESS RECOMMENDATIONS
============================================================

1. Payroll Cost Review
Highest payroll cost department: Marketing
Recommendation: Review staffing, workload and payroll expenses.

2. Overtime Management
Highest average overtime department: HR
Recommendation: Check workload distribution and overtime patterns.

3. Bonus Analysis
Highest average bonus department: IT
Recommendation: Review bonus criteria and performance policies.

4. Salary Fairness
```

Output

```
Gender
Female    74559.73
Male      74430.32
Name: Total_Salary, dtype: float64
```

Output

```
Recommendation: Compare salaries by job level, experience and performance.

5. Overall Recommendations
- Monitor department-wise payroll costs regularly.
- Investigate unusual salary and overtime patterns.
- Use transparent salary and bonus policies.
- Make cost decisions without compromising employee fairness.

Business recommendations generated successfully!
```

Code Cell 22

```
#20. Final Project Verification

print("=" * 50)
print("FINAL PROJECT VERIFICATION")
print("=" * 50)

print("Project: Payroll Cost Optimization & Salary Fairness Analyzer")
print("Total Rows:", df.shape[0])
print("Total Columns:", df.shape[1])
print("Missing Values:", df.isnull().sum().sum())
print("Duplicate Rows:", df.duplicated().sum())

df.to_csv("payroll_dataset.csv", index=False)

print("\nDataset exported successfully!")
print("File: payroll_dataset.csv")
print("Project review completed!")
```

Output

```
==================================================
FINAL PROJECT VERIFICATION
==================================================
Project: Payroll Cost Optimization & Salary Fairness Analyzer
Total Rows: 100000
Total Columns: 14
Missing Values: 0
Duplicate Rows: 0

Dataset exported successfully!
File: payroll_dataset.csv
Project review completed!
```